GATE-2003
BT: BIOTECHNOLOGY

Q. 1 - 10 Carry one mark each

1. Which of the following processes require energy?
   (a) ligation
   (b) transformation
   (c) restriction digestion
   (d) hybridization
2. To be a cloning vector, a plasmid does NOT require
   (a) an origin of replication
   (b) an antibiotic resistance marker
   (c) a restriction site
   (d) to have a high copy number
3. In animal cell cultures, the addition of serum to media is essential for providing
   (a) amino acids for protein synthesis
   (b) nucleotides for DNA synthesis
   (c) growth factors
   (d) all of these
4. In the course of cell cycle, the level of the protein cyclin abruptly falls during
   (a) G_1 phase (b) S phase
   (c) G_2 phase (d) M phase
5. Enzyme papain is used with success to
   (a) increase meat production
   (b) leaven bread
   (c) ripen papaya fruit
   (d) tenderize meat
6. Microbes bring about biological transformation of xenobiotic compounds by
   (a) degradation
   (b) conjugation
   (c) detoxification
   (d) all of these
7. Which one of the following reactions is used for the purpose of recycling enzymes inbioprocesses ?
   (a) Isomerization
   (b) Immobilization
   (c) Phosphorylation
   (d) Polymerization
8. The separation principle of dialysis used in the recovery of fermentation products is
   (a) diffusion
   (b) charge
   (c) turbulence
   (d) solubility
9. Identify the statement that is NOT correct
   (a) Penicillin fermentation is an aerobic process
   (b) Penicillin biosynthesis affected by phosphate concentration
   (c) Lysine stimulates penicillin synthesis
   (d) Penicillin production shows a distinct catabolite repression by glucose
10. For protoplast fusion to be successful in plant cells
    (a) fusion agents other than polyethylene glycol should be used
    (b) cell wall of the two strains of cells should not be damaged
    (c) DNA between the two cells should be compatible
    (d) osmolarity of the medium is not important

Q. 11 - 30 carry two marks each

11. A batch culture fermentation was being conducted with streptomyces rimosus. Analysis of samples collected indicated doubling of cell number per unit time. Youor inference would be that the culture is in the
    (a) lag phase
    (b) log phase
    (c) stationary phase
    (d) death phase
12. An animal cell line was transfected with DNA extracted from a tumorous tissue. Which one of the following will be diagnostic of its tumorous transformation ?
    (a) altered cell shape
    (b) contact inhibition
    (c) anchorage-independent cell division
    (d) increased duration of cell cycle

13. In order to identify the person who committed a crime, forensic experts will need to extract DNA from the tissue sample collected at the crime scene and conduct one of the following procedures for DNA finger-printing analysis

(a) cut the DNA and hybridize with specific micro-satellite probes

(b) cut the DNA and subclone the fragments

(c) determine the sequence of the subclones

(d) (b) followed by (c)

Answer Q. 14 – 15 based on the data given below :

Dr. Singh isolated a new 5-kb gene and wants to determine its sequence using a sequencer that can sequence upto 500 bases in a single reaction. Therefore, she decides to create subclones having suitable-size inserts for sequencing

14. Which one of the following will be the most appropriate restriction enzyme for this subcloning ?

(a) 8-bp cutter

(b) 6-bp cutter

(c) 3-bp cutter

(d) 5-bp cutter

15. To generate the minimum number of subclones needed for sequencing, what should be the size of the insert in these subclones ?

(a) 1000 bp

(b) 500 bp

(c) 250 bp

(d) 2000 bp

16. All of the following are true about DNA microarray technology except

(a) an electron microscope is used to gather data from the arrays

(b) the technology is used to assess transcription from multiple genes simultaneously

(c) the technology works best for organisms whose genome is completely sequenced

(d) the technology is derived from computer chip manufacture

17. You have cut the genome of a double-stranded viral genome with a restriction endonuclease and electrophoresed the products on an agarose gel. You observe only one band on the gel, equivalent to the size of the genome. This is because

(a) there are no introns in the genome

(b) the introns contain the recognition sites and have already been spliced out

(c) all of restriction fragments are too small to detect

(d) restriction endonucleases do not cut RNA, and this virus has an RNA genome

18. The restriction endonuclease Eco 52I recognizes the sequence C/GGCGG and cuts between the first C and the first G, indicated by the slash. DNA cut by which of the following enzymes (given with their recognition sequences and cut sites) could be cloned into a plasmid digested with Eco 52I ?

(a) EcoRI (G/AATTC)

(b) XmaIII (C/GGCGG)

(c) SmaI (CCC/GGG)

(d) SacII (CCGC/GG)

19. If bacterial cells are transformed with a mixture of linear and circular molecules resulting from a ligation reaction designed to produce a recombinant molecule

(a) no recombinant molecule will ever be detected

(b) both linear and circular molecules will replicate equally well

(c) none of the plasmids will express the antibiotic resistance gene located on the plasmid

(d) the circular molecules will be amplified by the cells

20. What is the primary purpose of neomycin in creating mice with knock-outs in gene X?

(a) neomycin selects for the survival of embryonic stem cells (ES) that have incorporated the mutant gene X anywhere in the genome

(b) neomycin selects for the survival of ES cells. that have incorporated the mutant gene in the place of the wild-type gene

(c) neomycin prevents *Candida* infection during ES cell culture that does not have gene X

(d) neomycin makes the gene X knock - out mice resistant to *Candida* infection

21. Match the industrial application of the following enzymes

P	Penicillinase	1. Pharmaceutical
Q	Pectinase	2. Leather
R	Trypsin	3. Wine
S	Rennin	2. Dairy

(a)	(b)	(c)	(d)
P-4	P-1	P-1	P-4
Q-3	Q-3	Q-2	Q-2
R-1	R-2	R-3	R-3
S-2	S-4	S-4	S-1

22. To optimize the bioreactor system, which one of the following conditions is least important for anaerobic fermentation ?
(a) culture agitation to maintain oxygen supply
(b) restriction of the entry of contaminating organisms
(c) control of parameters like pH and temperature
(d) maintenance of constant culture volume

23. Match the activity spectrum of the following antibiotics

P	Actinomycin D	1. Antifungal
Q	Daunorubicin	2. Antituberculosis
R	Rifamycin	3. Antitumor
S	Griseofulvin	4. Antiprotosoal

(a)	(b)	(c)	(d)
P-3	P-3	P-3	P-2
Q-4	Q-1	Q-1	Q-1
R-2	R-4	R-2	R-4
S-1	S-2	S-4	S-3

24. Autoclaves are routinely used in laboratories for sterilization. It acts by
(a) disrupting cell membranes
(b) denaturing proteins
(c) changing physically membrane lipids
(d) all of the above

25. All of the following produced by animal cells in culture and help the cells adhere to the culture dish except
(a) glycoproteins
(b) collagen
(c) phospholipase A
(d) hyaluronic acid

26. The following are useful to introduce genes into crop plants except
(a) Ti plasmid (b) Particle gun
(c) breeding (d) auxin

27. Power number, also called Newton's number, is defined as a dimensionless parameter relating to
(a) turbulent flow
(b) the relative velocity between the nutrient solution and individual cells
(c) energy required by the stirred reactors
(d) none of the above

28. The selection of the appropriate purification method in the product recovery after microbial fermentation depends on the
(a) nature and the stability of the end products produced
(b) type of the side products present
(c) degree of purification required
(d) all of these

29. Which of the following techniques is NOT ideal for immobilizing cell-free enzymes ?
(a) physical entrapment by encapsulation
(b) covalent chemical bonding to surface carriers
(c) physical bonding by flocculation
(d) covalent chemical bonding by cross-linking the precipitate

30. The full-length coding sequence of an eukaryotic gene was expressed in bacteria and the protein was purified. However, in the functional assay, no activity was detected for the purified protein. The reason could be
(a) the host bacteria produced an enzyme that inhibited the activity of the expressed eukaryotic protein
(b) the purified protein was contaminated with bacteria
(c) the host bacteria did not produce the essential co-factors
(d) no post-translational modification on the protein expressed in bacteria

ANSWERS

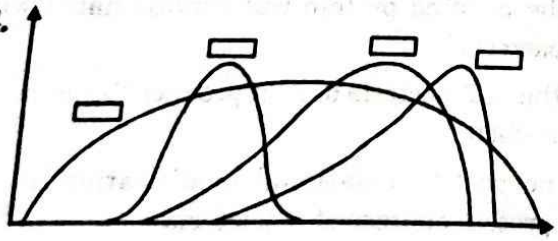
1. (a)	2. (b)	3. (d)	4. (d)	5. (d)	6. (d)	7. (b)	8. (a)	9. (c)	10. (c)
11. (b)	12. (c)	13. (a)	14. (c)	15. (b)	16. (a)	17. (d)	18. (b)	19. (d)	20. (a)
21. (b)	22. (a)	23. (a)	24. (d)	25. (c)	26. (d)	27. (c)	28. (d)	29. (d)	30. (a, d)

EXPLANATIONS

1. ligation is the ATP dependent procedure. The mechanism of DNA ligase is to form two covalent phosphodiester bonds between 3′ hydroxyl ends of one nucleotide, ("acceptor") with the 5′ phosphate end of another ("donor"). ATP is required for the ligase reaction.

2. for a cloning vector, origin of replication, a restriction site, high copy number and a selectable marker are required. A selectable marker can be of any type not necessarily to be an antibiotic resistance site.

3. • Serum provides the basic nutrients (both in the solution as well as bound to the proteins) for cells.
   - Serum provides several growth factors and hormones involved in growth promotion and specialized cell function.
   - It provides several binding proteins like albumin, transferrin, which can carry other molecules into the cell. For example: albumin carries lipids, vitamins, hormones, etc into cells.
   - It also supplies proteins, like fibronectin, which promote attachment of cells to the substrate. It also provides spreading factors that help the cells to spread out before they begin to divide.
   - It provides protease inhibitors which protect cells from preolysis.
   - It also provides minerals, like Na+, K+, Zn2+, Fe2+, etc.
   - It increases the viscosity of medium and thus, protects cells from mechanical damages during agitation of suspension cultures.
   - It also acts a buffer.

4. 

Papain has utility in breaking down tough meat fibres and has been used for thousands of years in its native South America. It is sold as a component of powdered meat tenderiser available in most super markets. Papain, in the form of such a powdered meat tenderizer, made into a paste with water, is also a home remedy treatment for, bee, and yellow jacket (w stings; mosquito bites; and possibly stir wounds, breaking down the protein toxins i venom.

6. microbes use replication procedure for xenol compound transformation. A xenobiotic chemical which is found in an organism but w is not normally produced or expected t present in it. It can also cover substances w are present in much higher concentrations are usual. Specifically, drugs such as antibi are xenobiotics in humans because the hu body does not produce them itself, nor are part of a normal diet. However, the t xenobiotics is very often used in the conte pollutants such as dioxins and polychlorin biphenyls and their effect on the biota, bec xenobiotics are understood as substances for to an entire biological system, i.e. artifi substances, which did not exist in nature be their synthesis by humans.

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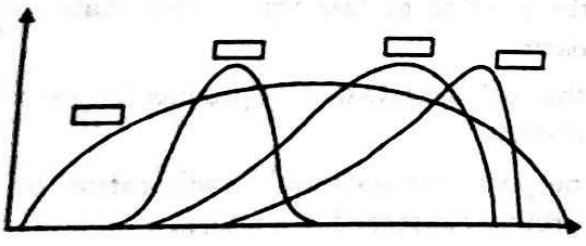
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11. Log phase shows the doubling in cell concentration with in a stipulated time unit.

12. Tumor tissue generally show anchor dependent nature. So Anchor independent test should be performed.

13. The vast majority of a human's DNA will match exactly that of any other human, making distinguishing between two people rather difficult. DNA finger printing uses a specific type of DNA sequence, known as a microsatellite, to make identification much easier. Microsatellites are short pieces of DNA which repeat many times in a given person's DNA. In a given area, microsatellites tend to be highly variable, making them ideal for DNA finger printing. By comparing a number of microsatellites in a given area, one can identify a person relatively easily.

14. Total gene size = 5000bp, Because sequencer can sequence upto 500 bases in a single reaction. Then the fragments are = 5000/500 = 10

Therefore $4^8 = 6536$, $4^6 = 4096$, $4^5 = 1024$, $4^3 = 64$

Only 3 – bp cutter will generate more than 10 fragments (5000/64 = 78). Rest other cutter will generate less than 10 fragments.

15. insert size should be 500bp

16. Array data is naked eye visible entity.

17. RE doesn't recognize RNA.

18. Check for cutting site specification. Cut by XmaIII can be cloned into a plasmid with Eco521.

19. Bacterial replication machinery is adapted for both linear and circular molecule in forms of genomic DNA and plasmid DNA but not equally.

20. Common approach to inactivate a selected gene is to insert an antibiotic resistance gene into the cloned gene of interest. This both inactivates the gene of interest and allows identification of ES cells that have been successfully transfected. In the engineered insertion construct, the target gene is disrupted with the neoR gene which confers neomycin resistance.

21. Beta-lactamases (penicillinase) are enzymes produced by some bacteria that provide resistance to beta-lactam antibiotics like penicillins, cephamycins and carbapenems. Beta-lactamase provides antibiotic resistance by breaking the antibiotic's' structure(a four-atom ring known as a beta-lactam). Through hydrolysis, the lactamase enzyme breaks the β-lactam ring open, deactivating the molecule's antibacterial properties.

Pectinases have also been used in wine production since the 1960s. The function of pectinase in brewing is twofold, first it helps breakdown the plant (typically fruit) material and so helps the extraction of flavours from the mash. Secondly the presence of pectin in finished wine causes a haze or slight cloudiness, Pectinase is used to break this down and so clear the wine.

One of the steps in leather industry is 'bating' which is done to make the leather soft and supple by using proteolytic bating enzymes of pancreatic or bacterial origin. Trypsin and alkaline proteases are commonly used.

Rennin is an enzyme generated in the stomachs of young mammals to help them digest milk. It is also the active component in rennet, a substance which is used to make cheeses and other dairy products such as junket.

22. anerobic fermentation doesn't need agitation to maintain oxygen supply.

23. *Griseofulvin* is an antifungal drug that is administered orally. It is used both in animals and in humans,to treat fungal infections of the skin (known as ringworm) and nails. It is derived from the mold *Penicillium griseofulvum*. *Daunorubicin* is chemotherapeutic of the anthracycline family that is given as a treatment for some types of cancer. It was initially isolated from *Streptomyces peucetius*.

The *rifamycins* are a group of antibiotics that are synthesized either naturally by the bacterium *Amycolatopsis mediterranei* or artificially. They are a subclass of the larger family Ansamycin. Rifamycins are particularly effective against mycobacteria, and are therefore used to treat tuberculosis, leprosy, and mycobacterium avium complex (MAC) infections.

24. Moist heat coagulates the proteins in any organism and this is aided by the water vapor that has a very high penetrating property, leading to their death. It also causes oxidative free radical damage. This can even, at high enough temperatures, kill prions.

25. A phospholipase is an enzyme that hydrolyzes phospholipids into fatty acids and other lipophilic substances. There are four major classes, termed A, B, C and D, distinguished by the type of reaction which they catalyse.

26. It's a hormone (auxin).

27. The *power number*, N_p is a commonly-used dimensionless number relating the resistance force to the intertia force.

The *power number* has different specifications according to the field of application e.g., for stirrers, the power number is defined as :

$$N_p = \frac{P}{\rho n^3 d^5} \text{ with}$$

P = power

ρ = fluid density

n = rotational speed

d = diameter of stirrer

28. The degree of protein purity required depends on the intended end use of the protein. For some applications, a crude extract is sufficient. However, for other uses, such as in foods and pharmaceuticals, a high level of purity is required. In order to achieve this, several protein purification methods are typically used, in a series of purification steps. Protein purification step also differs for intra-cellular and extra-cellular products.

29. Covalent chemical bonding by cross-linking the precipitate is not ideal for immobilizing cell-free enzymes.

30. Prokaryotes do not carry out the same kinds of post- translational modifications such as glycosylation, phosphorylation as eukaryotes do. This affects a protein's activity or stability, or at least its response to antibodies.